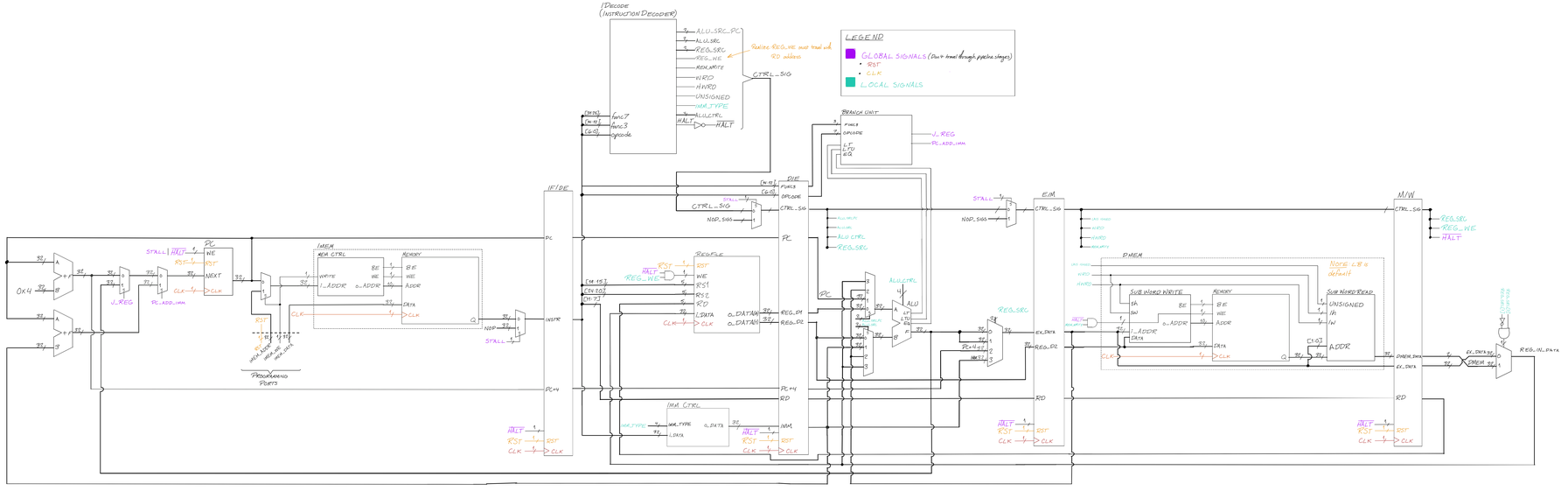

LEGEND
GLOBAL SIGNALS (pass through entire pipeline stages)
RST
CLK
LOCAL SIGNALS
IDECODE
(INSTRUCTION DECODER)
ALU_SRC_PC
ALU_SRC
REG_SRC
REG_WE
MEM_WRITE
HWRD
HWRD
UNSIGNED
IMM_TYPE
ALU_CTRL
HALT
Replace REG_WE once read with RD address
CTRL_SIG
func7
func3
opcode
BRANCH UNIT
FUNC3
OPCODE
LT
LTU
EQ
J_REG
PC_ADD_IMM
IF/DE
DE/E
E/M
M/W
PC
INSTR
PC+4
RD
REG_D1
REG_D2
IMEM
MEM CTRL
WRITE
I_ADDR
O_ADDR
MEMORY
BE
WE
ADDR
DATA
CLK
PROGRAMMING PORTS
REGFILE
RST
WE
RS1
RS2
RD
I_DATA
O_DATA1
O_DATA2
IMM. CTRL.
IMM_TYPE
I_DATA
O_DATA
NOP_SIGS
STALL
HALT
RST
CLK
ALU
ALU_CTRL
PC+4
REG_SRC
EX_DATA
DMEM
SUB WORD WRITE
SUB WORD READ
UNSIGNED
ADDR
Normal LW is default
DMEM_SWDATA
REG_IN_DATA
REG_SRC
REG_WE
HALT
0X4